

# Design of a Small Scale CFB Boiler Combustion Chamber for Laboratory Purposes

The [original paper](#) contains 15 sections, with 8 passages identified by our machine learning algorithms as central to this paper.

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## Paper Summary

### SUMMARY PASSAGE 1

#### Introduction

The main objective of this project is to enable the use of this technology in agricultural based country like Nepal where biomass waste could be used to power the boiler and obtain useful energy. This paper mainly concerns with the design (thermal) of a circulating CFB boiler combustion chamber for laboratory purpose and a very minimalistic simulation process to give anyone about the combustion taking place inside the boiler. The use of CFB boiler has many advantages over other conventional methods of steam generation as there is reduced NO<sub>x</sub> and SO<sub>x</sub> emission including the use of low-grade fuel to obtain high power output.

### SUMMARY PASSAGE 2

#### Design Process

The furnace of the boiler is designed by taking the standardized combustion process with excess air of 20%. The indirect method used for efficiency calculation is taken as a basis for the design of the required domain for the furnace. Specifying the resident time, using the stoichiometric relations and determining the environmental parameters the dimension of the furnace and minimum fluidization velocity were calculated.

SUMMARY PASSAGE 3

## Results And Discussion

The design of the furnace was the initial design step in the project and was carried out in MS Excel and as the design of the fuel hopper and cyclone were dependent on the design of furnace they were designed accordingly.

SUMMARY PASSAGE 4

## Figure1. Final Design

The above figure represents the final design of the complete CFB boiler combustion chamber (2) with hopper (1) and cyclone-syphon (3) assembly.

SUMMARY PASSAGE 5

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figure, the bed and temperature are increasing as soon as the rice husk particles are burning and finally obtained the maximum value of temperature after coal combustion is 1869K.

SUMMARY PASSAGE 6

## Co And Co<sub>2</sub> Mass Fraction

According to given contour plot Figure 3, CO mass fraction is zero at the lower end of the furnace this is because as the fuel and air are mixed combustion could not occurred at the starting time after certain time combustion of the rice husk occurred at the upper part of chamber and maximum value was found 0.14.

SUMMARY PASSAGE 7

## Pollutants (So<sub>2</sub> And No<sub>2</sub>) Mass Fraction

This is also required result for the CFB boiler. We know NO<sub>x</sub> emission is not occurred in the CFB boiler because of lower combustion temperature. NO<sub>x</sub> formation occurred at higher temperature that is more than 1200 °C. Here maximum fraction of NO<sub>x</sub> is 0.949 which is at the middle of the furnace this is because there is maximum temperature as shown in Figure 6.

## Mixture Fraction And Density

According to Figure 8 maximum density is at the inlet of the fuel because of small cross sectional area and minimum density is inside the furnace because air and fuel are disturbed all over the furnace. Hence, the objective of designing a fluidized bed combustion chamber along with is respective fuel hopper and cyclone separator has been fully accomplished.